

① Energy survey +10

② Graded on completion +10

③ a) Instantaneous +2 → includes flow rate +4

$$\frac{dM_{\text{barrel}}}{dt} = \dot{m}_{\text{in}} - \underbrace{\dot{m}_{\text{out}}}_{\substack{\rightarrow 0 \\ \text{no leaking}}} + \underbrace{\dot{m}_{\text{gen}}}_{\substack{\rightarrow 0 \\ \text{no mass generated}}} \rightarrow$$

$$\boxed{\frac{dM_{\text{water}}}{dt} = \dot{m}_{\text{in}}} +2$$

b) Instantaneous +2 → heating rate

$$\frac{dE_{\text{H}_2\text{O}}}{dt} = \underbrace{\dot{E}_{\text{in}}}_{\substack{\downarrow \\ \dot{Q}_{\text{heat}}}} - \underbrace{\dot{E}_{\text{out}}}_{\substack{\downarrow \\ \dot{Q}_{\text{loss}}}} + \underbrace{\dot{E}_{\text{gen}}}_{\substack{\rightarrow 0 \\ \text{no energy generated}}} \Rightarrow$$

$$\boxed{\frac{dE_{\text{H}_2\text{O}}}{dt} = \dot{Q}_{\text{heat}} - \dot{Q}_{\text{loss}}} +2$$

c) Finite time +2 → 1-minute duration

$$\Delta E_{\text{H}_2\text{O}} = \underbrace{E_{\text{in}}}_{\substack{\downarrow \\ \dot{Q}_{\text{in}}}} - \underbrace{E_{\text{out}}}_{\substack{\rightarrow 0 \\ \text{no loss}}} + \underbrace{E_{\text{gen}}}_{\substack{\rightarrow 0 \\ \text{no energy gen}}} \Rightarrow$$

$$\boxed{\Delta E_{\text{H}_2\text{O}} = Q_{\text{in}}} +2$$

d) Instantaneous +2 → flow rate

$$\frac{dE_{\text{pipe}}}{dt} = \underbrace{\dot{E}_{\text{in}}}_{\substack{\downarrow \\ \dot{m}_{\text{in}} + \dot{W}_{\text{comp}}}} - \underbrace{\dot{E}_{\text{out}}}_{\substack{\downarrow \\ \dot{m}_{\text{out}}}} + \underbrace{\dot{E}_{\text{gen}}}_{\substack{\rightarrow 0 \\ \text{no energy gen}}} \Rightarrow$$

$$\boxed{\frac{dE_{\text{pipe}}}{dt} = \dot{m}_{\text{in}} + \dot{W}_{\text{comp}} - \dot{m}_{\text{out}}} +2$$

c) Finite time ⁺² → takes place over the course of a day

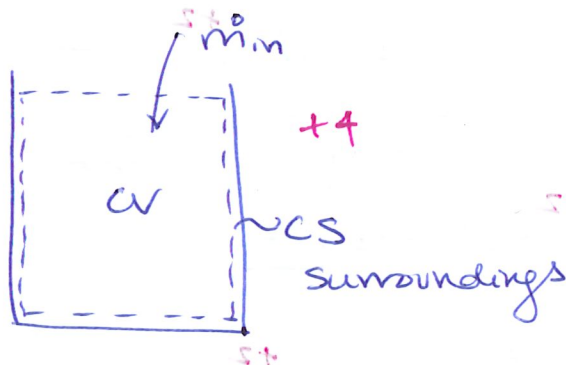
$$\rightarrow \Delta E_{\text{battery}} = E_{\text{in}} - E_{\text{out}} + E_{\text{gen}} \quad \Rightarrow \quad \Delta E_{\text{battery}} = E_{\text{solar}} - Q_{\text{loss}} - E_{\text{out}}$$

$\downarrow$ $\downarrow$ $\swarrow$ ⁺²
 E_{solar} $Q_{\text{loss}} + E_{\text{chase}}$ $\rightarrow$ no energy gen
C/T

④ a) Rain barrel

→ control volume ⁺¹ ⇒ flow in ⁺¹

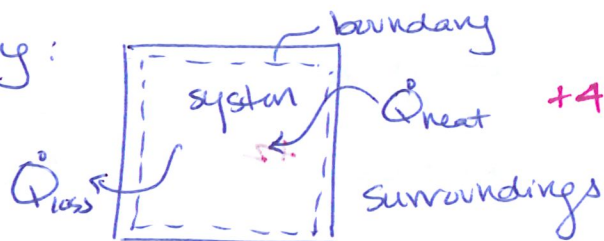
→ drawing:



b) Mr. Coffee

→ system ⁺¹ ⇒ constant mass of water ⁺¹

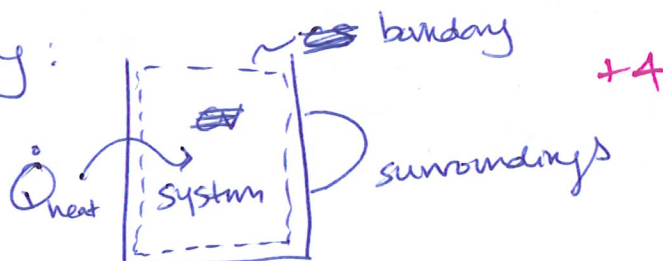
→ drawing:



c) Coffee cup

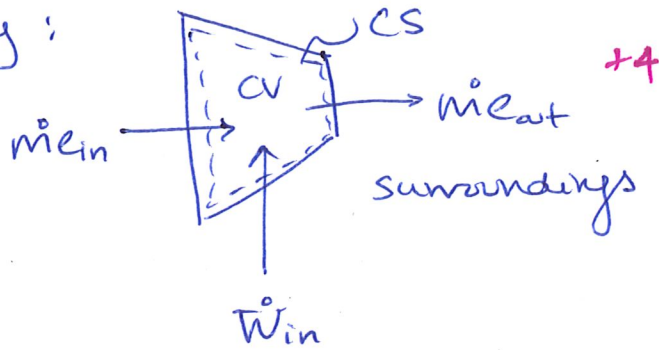
→ ~~const~~ system ⁺¹ ⇒ constant mass of water ⁺¹

→ drawing:



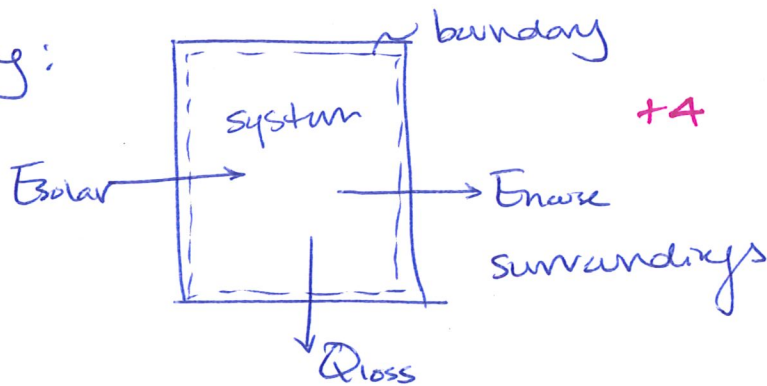
d) control volume ⁺¹ → flow through the pipe ⁺¹

Drawing:



e) system ⁺¹ → battery is constant mass ⁺¹

Drawing:



- ⑤
- Pressure - Pascals [Pa] ⁺¹
 - Temperature - kelvin [K] ⁺¹
 - Volume - Meters cubed [m^3] ⁺¹
 - Energy - Joules [J] ⁺¹
 - Power - Watts [W] ⁺¹
 - Length - meters [m] ⁺¹
 - Time - seconds [s] ⁺¹
 - Mass - kilograms [kg] ⁺¹
 - Velocity - meters per second [m/s] ⁺¹
 - Density - kilograms per meters cubed [kg/m^3] ⁺¹

6 Based on completion +10